

Case Reference: CASE-TXN-4F5385C7
Transaction ID: TXN-4F5385C7
Timestamp: 2026-08-16 15:40:39 UTC
Amount: **\$14,900.00 USD**
Action Taken: STEP_UP_AUTH

RISK LEVEL: MEDIUM
30.5/100

Status: Transaction restricted; dossier preserved
for compliance and audit trail.

1. Multi-Factor Risk Model Diagnostics

Diagnostic Factor	Raw Metric	Factor Weight	Normalized Impact
Supervised Machine Learning (XGBoost)	0.1% Fraud Probability	65%	0.0 / 65.0
Unsupervised Anomaly (Isolation Forest)	100.0 Outlier Index	25%	25.0 / 25.0
Domain Heuristics & Rules	0 Rule(s) Triggered	10%	1.0 / 10.0
CALIBRATED COMPOSITE RISK SCORE	MEDIUM SEVERITY	100%	30.5 / 100.0

2. Explainable AI (SHAP) Forensic Feature Attribution

Primary Behavioral Triggers Identified:

- Latent identity / card-use deviation (V14 component)
- Transaction amount pattern (\$14,900.00)
- Multi-vector behavioral anomaly score

Feature Name	Observed Value	SHAP Attribution (Delta)	Risk Direction
V14	-5.24	+0.3640	Elevates Fraud Probability
V12	-3.88	+0.2810	Elevates Fraud Probability
Amount_Deviation_Z	2.45	+0.1650	Elevates Fraud Probability

3. Fraud Analyst Investigation Notes & Audit Signature

Investigator: Lead Fraud Analyst #8142

Findings: Explorer review: Action recommendation STEP_UP_AUTH

Digital Audit Checksum (SHA-256):

86a71078c8c210a119b9cbe9ca08f77626001a5a78306e0bce2667620295d643

Verified Fraud Risk
Engine

COMPLIANCE CERTIFIED